Time Series Data and Recent Imputation Techniques for Missing Data: A Review

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***Abstract*— The development of multisensory systems and the ongoing application of data collection technologies have both contributed to the explosion of time series data. However, due to many factors, undesirable missing data points are often encountered. The inability to analyse and model the missing data greatly hinders categorization and forecasting activities. Traditional techniques for processing time series data frequently add bias and make significant assumptions about the underlying data creation process, which can result in inaccurate development of prediction or classification models. The characteristic of the time series data needs to be well understood before applying the correct approach for imputation. This study aims to brief the types of time series data, and missing data mechanisms and also reviews several approaches to filling data gaps that are convenient for time series data. The review highlights current approaches in handling missing values at the data pre-processing stage for univariate and multivariate time series data together with the methods used to evaluate the performance of the imputation approach. It includes some advantages and drawbacks of these approaches practically. The results provide information which can be used to further develop a new imputation approach.**

Keywords*— *data imputation, deep learning, machine learning, sparse data, univariate time series data

I. Introduction

Time series data is very valuable in data mining and analysis, but it often comes with the issue of data that is partially sparse that often occurs in real-world datasets in many domains. This prevalent problem often impairs the performance of data analytics by giving inaccurate and biased results. Missing data points may occur due to five (5) main reasons, incomplete data entry [1]–[3][4], errors in measurement [5], equipment malfunctions, communication and transmission issues [6], [7], or weather conditions.

The imputation of time series is one of the most significant activities involved in the pre-processing stage. The quality and precision of this process will directly impact the quality of the time series estimated. It is a recurring problem in data-driven modelling; hence, missing values must be imputed from raw data to enhance accuracy in time series analysis.

Through investigation and analysis of freshly presented approaches for data imputation, significant improvements in terms of accuracy, performance, and required time can be uncovered. The aim of this study is to provide a reference of 2019 to 2022 publications for researchers to familiarise themselves with the latest techniques used to impute missing values focusing on time series data. The main contributions of this article are summarized as follows;

i. It gives enlightenment on the types of time series data.
ii. It explains the types of missing data.
iii. It highlights emerging methods used in filing missing data for time series data.
iv. It summarized the imputation performance measurement approaches that have been implemented in previous work.

The rest of this paper is organised into five (5) main sections, beginning in Section II with a synopsis of the type of time series data in section A. Section B provides a description of time series components. It continues with section C explaining the categories of missing data. The body of the review is provided in Section III, which is then clustered into subsections following the types of methods used for univariate time series (UTS) data and multivariate time series (MTS) data imputation. Section IV is the summary of methods used to measure the performance of imputation and finally, Section V explains the findings as well as highlights the top methods that are currently used for time series data imputation. It follows with the areas of ongoing research and some recommendations for methods applied in this field.

II. Time Series Data

A time series data is a collection of information produced from continuous measurement over time. It is a collection of observations obtained through constant monitoring throughout time. As time is a component of all observable phenomena, time series data are ubiquitous allowing us to

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develop smarter systems, increase the efficiency of nearly all operations, and forecast future events in real time. The ability to collect and analyse time series data is becoming increasingly important in the commercial world and as a driver of innovation in the tech sector.

A. *Types of Time Series Data*

In most cases, it is expected that time series are created at regularly spaced intervals of time; hence, these time series are referred to as regular time series. A timestamp may be explicitly included in the data, or it may be inferred from the intervals at which the data is generated. Either way, the data may be used. The inverse of a regular time series is called an irregular time series. A temporal sequence is followed by the data in the time series, although the measurements may not occur at consistent and regular intervals of time across the whole series. For instance, seismic data are frequently collected irregularly and sparsely along the spatial coordinates. In the domain of temporal databases, the emphasis was placed on data that was related to a certain period, or that is valid at a certain time frame [8]. It may not valid across the year, just valid at a specific time duration. Temporal databases are stored in temporal order, and the analysis operations must take into account the time series as a whole rather than the values stored at individual time points. A different process is required to handle regular and irregular time series data.

Other than that, A time series dataset can have one or more variables being measured. Depending on how many variables are in a time series and how they relate to each other, time series data can be put into one of three (3) groups; univariate time series (UTS), multivariate time series (MTS) or multiple and summarized in the chart of Fig. 1.:

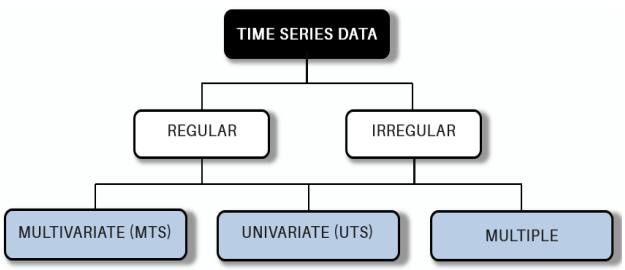

Fig. 1. Summary of time series data.

i. A UTS data comprised of repeated measurements of a single variable over the course of time.
ii. An MTS is comprised of the continuous measurements of multiple connected variables collected over the course of time. A bivariate time series contains two variables which are also considered MTS data. In MTS, each of these variables is connected to the others in some way.
iii. Multiple time series data contains measurements of multiple separate entities over the same time period. It can also be said of a dataset containing multiple univariate time series data.

B. *Types of Time Series Components*

Time series data have four components which are trend, seasonal, cyclical and random change:

i. Trend component: The data will show a trend of a peak, ascending and descending over a long-term measurement and monitoring. In most cases, it is a one-time occurrence that is followed by no more traces of itself.
ii. Seasonal component: The component goes through changes that are predictable and consistent and that occur at the same time each year. intra-interval fluctuations are taken into consideration, which indicates that there is a repeated pattern over a period of one year. For example, it could a pattern within a week, a month, or a year. These peaks and valleys can be found at different points within the time series. The term "seasonal" refers to any change or trend that can be predicted and that occurs or reoccurs at regular intervals over the course of a year.
iii. Cyclical component: This data is similar to data of seasonal components just the measurement and monitoring time is more than a year.
iv. Random change component: This component has an unpredictable pattern. It does not show a trend of a peak and low and fluctuation over a time period.

C. *Categories of Missing Data*

Data measurements are conducted in many different conditions, and having missing data is naturally unavoidable. Missing data can be categorized into three (3) types of so-called “missing mechanism”; Missing completely at random (MCAR), Missing at random (MAR) and Not missing at random (NMAR) [9], [10]. The illustration of the missingness mechanism is available in Fig. 2.

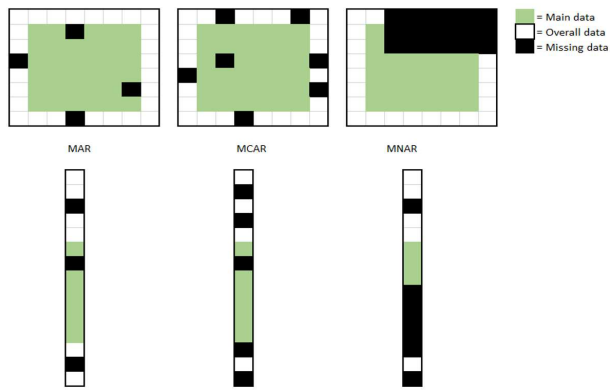

Fig. 2. Illustration of data missingness mechanism.

i. Missing at Random (MAR): The proclivity for a data point to be missing is unrelated to the missing data but is connected to some of the observed data.
ii. Missing Completely at Random (MCAR): The absence of a particular value has nothing to do with its prospective value or the values of other variables.
iii. Missing not at Random (MNAR): Two possible explanations are that the missing value is dependent on the hypothetical value or that the missing value is dependent on the value of another variable.

For time series data, it is always better to impute to have continuous data available for use. Continuous data sets of the same shape and resolution are often required for accurate model development and validation, this is both a hurdle and an incentive for study. Several approaches have been

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proposed to fix missing values according to their missingness mechanism from the conventional one to the more modern ways to handle missing values.

III. Recent Imputation Methods

A substantial study has been conducted to examine the various processes of missing data points and the appropriate approaches for dealing with diverse data types. An extensive amount of research has been published on the imputation of missing data to date. The simplest and most common methods used in imputing missing data are mean imputation such as using mean, mode or median data to replace the missing data points. Another is interpolation either linear, polynomial or spline interpolation and lastly is the Last Observation Carried Forward (LOCF) [11]–[13]. However, these methods are not suitable to handle multiple types of data gaps and especially large data gaps. It is most applicable for a few consecutive missing values or short data gaps. Thus, there are few studies on classical methods to impute data gaps using Auto-Regressive Integrated Moving Average (ARIMA) algorithms. [14], [15] and Hotdeck [1], [16], taking into account the statistical features of the analysis.

The processes for imputation of missing values can be either a single imputation, which merely offers one possible response to fill in the missing value or multiple imputations. Single imputation includes the likes of mean imputation, deck operations, regression, expectation-maximization, and deletion. While single imputation is a one-step process, multiple imputations are a multiple stages process that includes imputation, analysis, and merging. It generates a large number of complete datasets by repeatedly attempting to fill in the blanks for the missing values.

Recently, proposed methods for imputing time series data that result in more precise imputed data have been used to replace and improve on traditional and empirical approaches to solve the issue of missing values more effectively throughout the previous decade. The performance of data imputation is highly dependent on the selection of an appropriate imputation method for the type of missing data. With that, in this work, the imputation method is categorised according to these two (2) data types, UTS and MTS.

A. Imputation for Univariate Time Series Data (UTS)

Filling missing values in UTS data, a one-dimensional value is an exceptionally challenging job especially if it is non-linear and missing at random types of data. The available data could be very limited to be used for analysis and modelling. Most of the available imputation methods discovered are catered for MTS data. However, this section shares the available studies on UTS data, a single time-dependent variable imputation procedure which can be useful for future work.

One of the most used methods is the k-nearest neighbour's algorithm (kNN). Some improved and modified kNN have been done for many different fields. Dubey A. and Rasool A. have used top k-NN for imputing micro-array gene expressed data in bioinformatics with few local imputation methods before the top kNN [17]. It uses graph similarity in conjunction with the K-means clustering technique to provide an approximate of the incomplete data that proceed with the top kNN technique and the weighted function by cluster is put together. It applies to missing datasets ranging from 0.02 % to 48.05%. While Anibal Flores et al [18] on the other hand proposed two algorithms improved from kNN for short gaps in meteorological time series datasets within the UTS algorithms. The first is the Local Average of Nearest Neighbors (LANN) algorithm is based on the average of the two nearest neighbours, while LANN+ is based on the mean of the four (4) nearest neighbours to the missing value or NA.

A single index model was used by N. Ling et al [19] to reduce the complexity of the explanatory variable and escape the curse of dimensionality in multivariate regression. They build this model's estimators using the kernel approach and establish the uniform almost complete convergence rate and asymptotic distribution under some generic environments. On the other hand, Denitsa Grigorova et al [20] highlight two general approaches in multiple imputations (MI), one is by taking incomplete data from a multivariate distribution for all inadequate variables at the same time and another is multivariate imputation by chained equations (MICE) uses a series of univariate conditional distributions to estimate the missing data by taking the univariate density based on the other variables in the model. Two methods that focus on imputing large gaps of missing univariate data are Higher-Order Markov Chains [21] which uses a univariate imputation method with a multivariate imputation approach and another one is (DTWBI) used by T.T.H Phan et al [22] by finding the similar sub-sequence data within a window period before the gap for imputation.

A proposed method by Yuehua Liu et al [23] on how to handle larger data gaps was applied to IoT data of UTS. To compute the most accurate value for the missing points, the study used several segmented sparse iterations called It-MS-STLecImp. The gap was initially split into various sections to begin, then continue with gap reformation and gap synthesis. A different category of univariate data type is seasonal data type where Zhou Y. et al [24] and J. Park et al [25] have studies on suitable methods to impute them. Zhou Y. highlights more on imputing temporal dynamic time series data. Zhou Y. also highlighted that recurrent neural networks (RNN) [26] would only be suitable for short gaps. Park et al, use deep learning (DL) with a multilayer perceptron (MLP) approach for long consecutive missing data points.

A Machine Learning-based Univariate Time Series Imputation (MLBUI) method experimented with by T. Phan [27] turns a single-variable time series into a multivariable and uses all of the available values of the single variable to predict values that are missing. A new single imputation method known as "First five last three logistic regression imputation" (FTLRI) is introduced in order to obtain a more effective imputation of discrete missing data in air quality measurement. This technique incorporates the conventional regression models with the "first five & last three," which can explain relationships between the time and attribute [28].

It is observed that their empirical method is still applied for UTS data even though machine learning (ML) and DL approach is recently being added to the pool of imputation methods. Next, we look at imputation methods applied to MTS data.

B. Imputation for Multivariate Time Series Data (MTS)

One of the methods used to impute MTS data is the MICE to impute missing values in spatiotemporal data. Combining MICE with other imputation methods done by R.

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Wu [29] improved the accuracy of long consecutive missing data imputation by 23%. A few other studies applying MICE is Y. Luo [30] who discussed that the vast majority of clinical datasets are longitudinal by design and often have missing data yet, the majority of imputation algorithms that are commonly employed today do not directly handle longitudinal clinical time series. MICE is also used for filling the gaps in copper prices [31] and provides a very low measurement of RMSE and MAPE.

Regression imputation for MTS has been implemented using supervised kNN by [32] comparing it with super vector regression (SVR) and general regression neural network (GRNN) to impute missing ground electromagnetism data. It is discovered that SVR improved the missing data points by up to 80%. The Random Forest (RF) algorithm is also being used for MTS data using an average of the respective full values. The algorithm then repeatedly improves the missing data imputation by making use of proximity data points [33]. An improvised RF by Yu Gu et al [34] was done where in order to provide accurate projections of new data, the imputation of each tree was collected and then averaged. A natural approach to determining which factors are most important can be taken with the use of RF. The long short-term memory (LSTM) approach used by [7], [35] discovered that the method is useful for a variety of different missing patterns and delivers more accurate forecasts than the other techniques that are currently available.

In summary, MTS data are famously known to use MICE, kNN, RF and other DL approach such as LSTM to handle missing data. A summary of the collected studies is tabulated in Table 1.

TABLE I. RECENT METHODS USED FOR DATA IMPUTATION

Time Series Data Type	Methods	Ref.
Univariate (UTS)	k-Nearest Neighbour (kNN)	[17]
	Local Average of Nearest Neighbors (LANN)	[36] [37][18]
	Functional Single Index Regression (FSIRM)	[19]
	Multiple Imputation By Chained Equations (MICE)	[20]
	Higher-Order Markov Chains	[21]
	Dynamic Time Warping (DTWBI)	[22]
	Machine Learning-Based Univariate Time Series Imputation (MLBUI)	[27]
	First Five Last Three Logistic Regression Imputation (FLTRI)	[28]
Multivariate (MTS)	Multiple Imputation By Chained Equations (MICE)	[29], [30], [38], [39],[31]
	Imputation by Feature Importance	[40]
	Supervised k-Nearest Neighbour (kNN)	[32]
	Random Forest (RF)	[34], [33]
	Long Short-Term Memory	[35]

IV. IMPUTATION PERFORMANCE MEASUREMENT USED

In order to evaluate the performance of the series of imputed data, multiple methods have been used and compare their performance with other available existing or innovative methods. Of all the imputation techniques summarized, 70% of them used root mean squared error (RMSE) as the performance measurement, together with one or two other methods such as absolute percentage error (MAPE), mean absolute error (MAE), symmetric mean absolute percentage error (SMAPE), mean squared log error (MSLE), standard deviation (SD), and percent bias (PB).

V. CONCLUSION

The imputation method for time series data can be clustered into two groups; UTS and MTS missing data. In the case of UTS, the methods that can be utilised include, but are not limited to, univariate time series algorithms. Famously methods for univariate time series data are LANN, MICE, kNN and regression. While MTS, mostly used the approach of MICE, kNN and RF. However, a lot of methods can be explored by combining and innovating the methods to evaluate the most effective approach for a specific data type for a specific field. In general, most of the approaches are using regression in order to identify the pattern of the data and estimate the new data to fill in the existing gap. RMSE is a must measurement to be used to evaluate the accuracy of imputed data and the performance of the methods used.

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